Embedded systems exp Exp 25

Program

```
#include <systemc.h>

// Producer Module
SC_MODULE(Producer)
{
    sc_fifo_out<int> out;

    void produce()
    {
        for (int i = 1; i <= 5; i++)
        {
            cout << "Produced: " << i << endl;

            out.write(i);

            wait(1, SC_SEC);
        }
    }

    SC_CTOR(Producer)
    {
        SC_THREAD(produce);
    }
};

// Consumer Module
SC_MODULE(Consumer)
{
    sc_fifo_in<int> in;

    void consume()
```

```
    {
        int value;

        while (true)
        {
            in.read(value);

            cout << "Consumed: " << value << endl;

            wait(1, SC_SEC);
        }
    }

    SC_CTOR(Consumer)
    {
        SC_THREAD(consume);
    }
};

int sc_main(int argc, char* argv[])
{
    sc_fifo<int> fifo(5);

    Producer producer("Producer");
    Consumer consumer("Consumer");

    producer.out(fifo);
    consumer.in(fifo);

    sc_start(10, SC_SEC);

    return 0;
}
```

OUTPUT

```
Log   Share
        ALL RIGHTS RESERVED
Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
Done
```